Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Inventory Rescue

a. Team name: Fantastic Four
b. Team leader name: Ashish Kumar Singh
c. Problem Statement: Growing food & nutrition brands lose lakhs a month to stockouts and expired stock because supplier delays and demand spikes arrive as unstructured messages, and limited cash forces hard buy / expedite / discount trade-offs.

Brief about the idea

"With ₹10 lakh available this month, what should we buy, expedite, discount or defer, and when?"

The problem

- Mid-size and large D2C / FMCG brands (₹50 crore+ a year) still plan rescues on spreadsheets, email and WhatsApp.
- Supplier delays, influencer-driven spikes and expiring lots arrive as unstructured messages.
- Cash is tight, so hero SKUs stock out while slow movers expire and get written off.

Our solution

- Generative AI reads the messages (English, Hindi, Hinglish), turns text-only news into forecast adjustments and answers "why / what if" questions with planner tools.
- A deterministic simulator and optimiser decide quantities and money; every LLM output is validated.
- The operator confirms signals, picks when to spend and exports POs.

The outcome

- One ranked action list within budget, plus four ways to time the spend (commit now, staggered, reserve, lean).
- Demo brand (₹4.3 crore revenue a month): **₹3.8 lakh / month** lower simulated loss than a static reorder rule and **₹1.8 lakh** than a tuned one, better in **120 of 120** simulated futures.
- Keeping 30% in reserve for weekly reviews: a further **₹0.9–1.1 lakh** in 9–10 of 10 futures.

Target users: starting with supply-chain and operations teams of mid-size to large FMCG / D2C brands across JAPAC; the engine is product-agnostic. All values are simulated on seeded data and will be validated in pilots.

Opportunities

How is it different from existing ideas?

- **Inventory apps / ERPs** track stock but use static reorder points and stated lead times, and ignore supplier or campaign messages.
- **Enterprise demand planners** forecast well but take months to roll out and still ignore the inbox; the rescue decisions happen in spreadsheets.
- **Generic AI chatbots** give advice without arithmetic and can invent numbers.
- **Ours** pairs LLM understanding with budget-aware optimisation and a built-in proof of value.

How will it solve the problem?

- Turns messages into typed events with verbatim evidence; a human confirms.
- Hybrid forecasts: a statistical base, with the LLM tuning event effects that only exist as text.
- Supplier delay history gives early / typical / late arrival cases.
- Picks the best action mix under one cash budget, tests it on 120 simulated futures and shows when to spend.

USP of the proposed solution

- **One cash budget** across buy, expedite, markdown and defer, and over time (staggered buys).
- **Trust by design:** the LLM proposes, code validates; every number in an answer is checked.
- **Measured, not claimed:** extraction gold set, forecast backtest, Monte Carlo, rule baselines.
- **Built on Google Cloud:** Gemini on Vertex AI, Cloud Run, Firebase Hosting, Firestore, Cloud SQL.

Why now: LLMs can finally read messy supplier and campaign messages well enough, when wrapped in validation, to feed real planning maths, and run it on pay-per-use cloud for a fraction of what it saves.

List of features offered by the solution

1 Signal Inbox

Emails, WhatsApp notes, Hindi/Hinglish → typed delay, spike, drop and expiry events with verbatim quotes. A human confirms each one.

2 Hybrid forecasts

Statistical base (Holt-Winters / Croston). The LLM tunes event effects within bounds, including news that exists only as text.

3 Planning assistant

Ask “why” or “what if” in plain language. The model calls 10 read-only planner tools; every number is checked.

4 Grounded explanations

Plan narrative and what-if parsing that may only use numbers and IDs from the plan; else a labelled template.

5 Supply & service level

Supplier delay cases (early / typical / late); hero SKUs carry a goodwill penalty per unit short.

6 Budgeted planner

Buy, expedite, markdown (with cannibalisation) or defer; an exact knapsack picks the best set under the cash budget.

7 Plan options

Commit now, staggered buys, keep a reserve or lean, compared with weekly re-planning in calm and shocked futures.

8 Robustness & proof

120 Monte Carlo futures vs static and tuned reorder rules; budget frontier; forecast backtest.

9 Overrides & plan diff

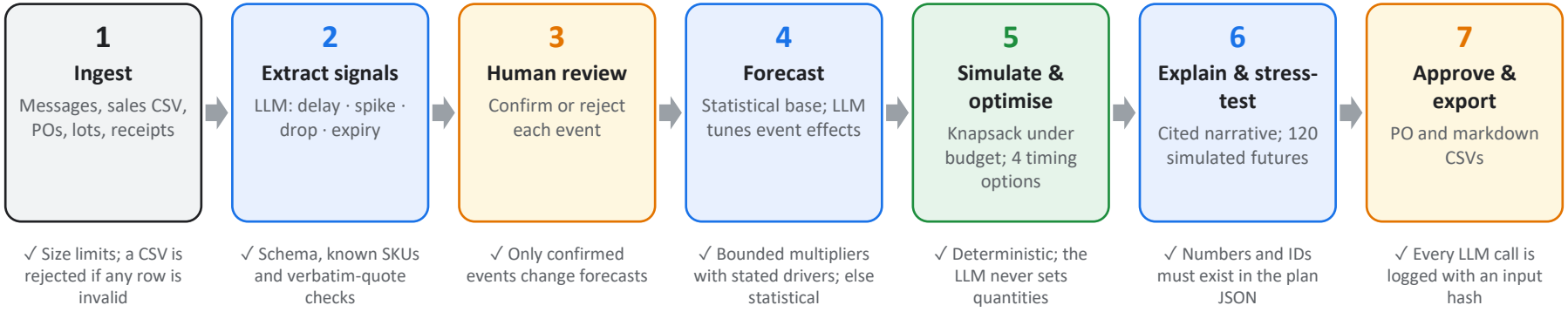
Edit quantities or skip lines; the plan is re-scored within budget and each re-plan lists what changed.

10 Approve & export

Alerts for stockout, expiry, overstock and suppliers; approve to export PO and markdown CSVs.

Blue = generative AI · Green = deterministic engine · Orange = operator workflow

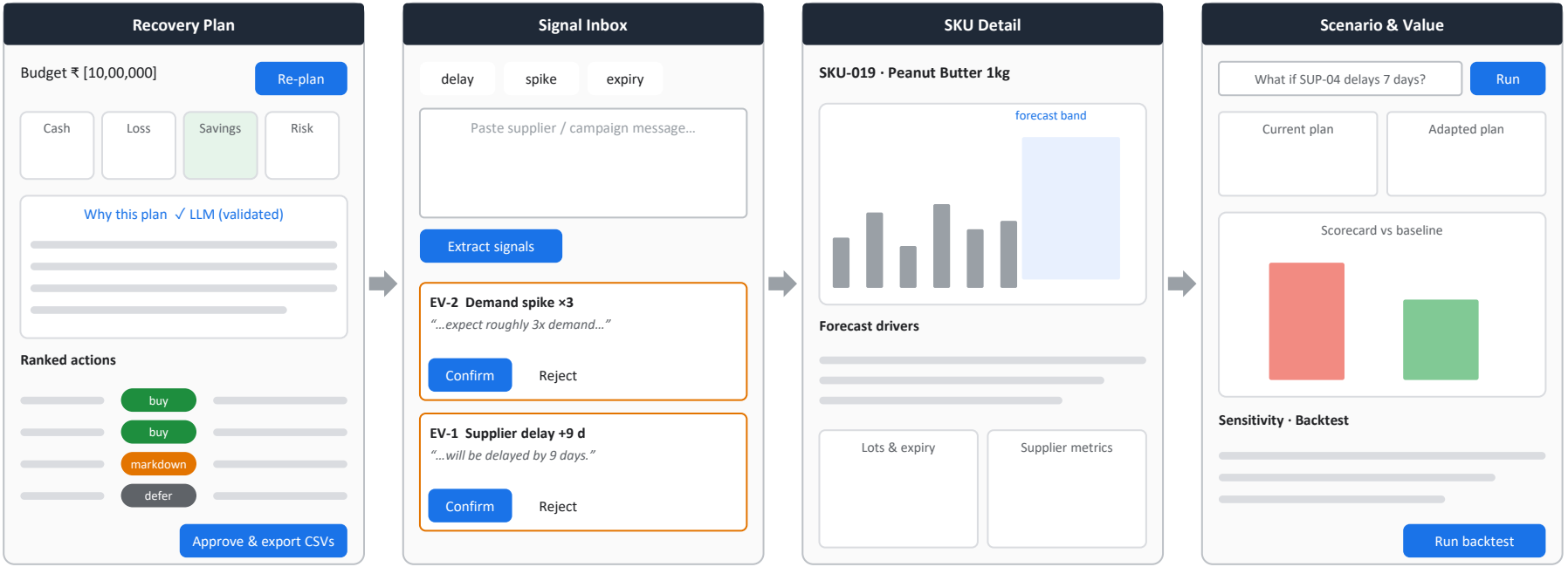
Process flow diagram



↻ **Weekly reviews (days 0 / 7 / 14 / 21):** realised sales re-forecast → remaining and scheduled cash re-planned → staggered tranches re-confirmed, resized or redirected

Ask the assistant: “What if SUP-04 slips another week?” → the model calls read-only planner and simulator tools → answer with a tool trace; every number checked against tool results

Wireframes / mock diagrams of the proposed solution



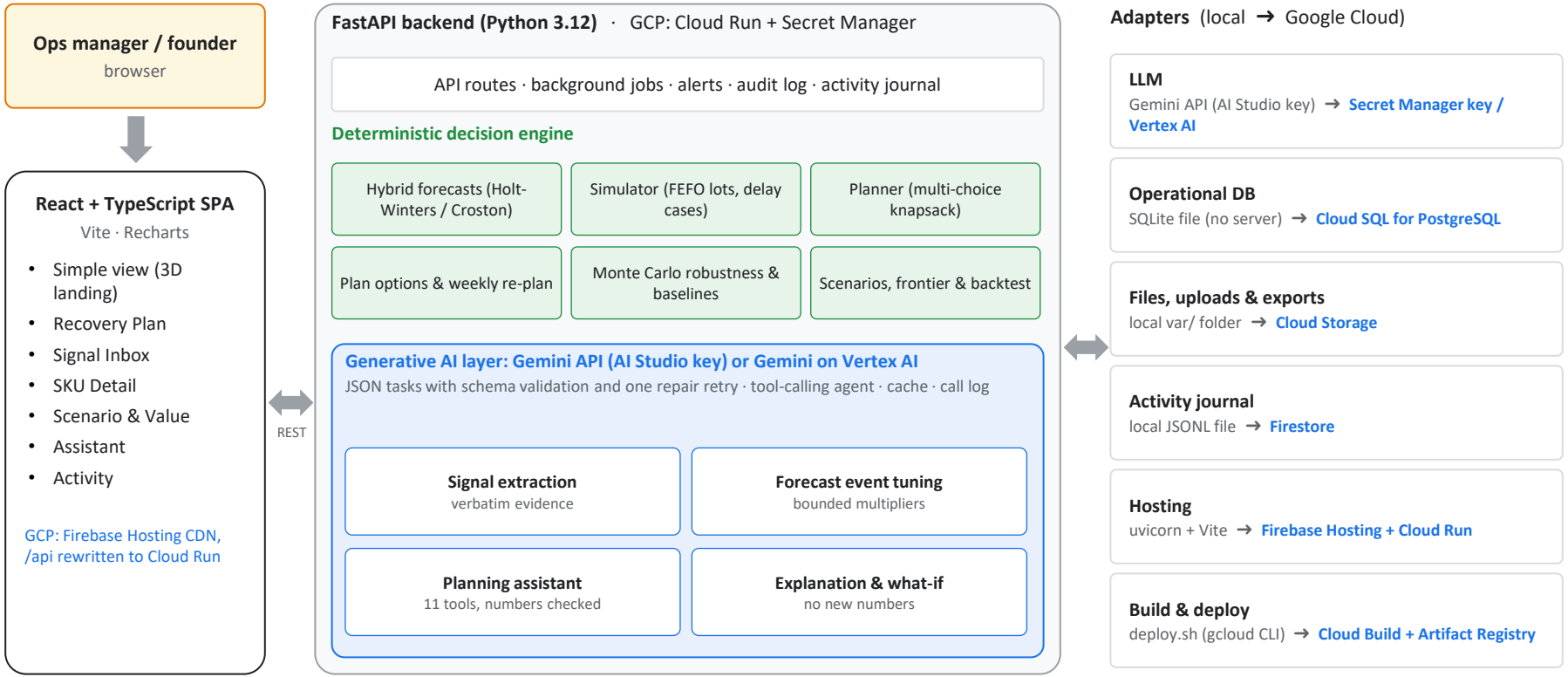
1. Decide & approve

2. Capture signals

3. Inspect a SKU

4. Stress-test value

Architecture diagram of the proposed solution



Technologies to be used in the solution

Layer	Prototype (built and running)	Google Cloud (production)
Frontend	React 19, TypeScript, Vite, Recharts, three.js	Firebase Hosting (CDN), /api rewritten to Cloud Run
Backend API	Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2, Uvicorn	Cloud Run, Secret Manager
Generative AI	google-genai, gemini-3.5-flash-lite via the Gemini API or Vertex AI; validated JSON tasks; tool-calling assistant	Gemini API (key in Secret Manager) or Gemini on Vertex AI
Decision engine	Holt-Winters and Croston/SBA in NumPy; stochastic simulator; exact knapsack	Same code in Cloud Run (CPU always allocated)
Data & files	SQLite (no database server); local folder for uploads, exports, forecast cache	Cloud SQL for PostgreSQL, Cloud Storage, Firestore (activity journal)
Evaluation	Extraction gold set, rolling backtest, Monte Carlo, rule baselines	Same jobs; CSV exports ready for BigQuery
Quality & delivery	pytest (162 tests), oxlint, strict TypeScript, LLM call log	Cloud Build → Artifact Registry → Cloud Run

Estimated implementation cost

Component (one brand, production sizing)	US$ / month
Cloud Run: API + UI, 2 vCPU / 2 GiB, one always-on instance	90 – 105
Cloud SQL for PostgreSQL 16: db-g1-small, 10 GB	25 – 40
Gemini 3.5 Flash-Lite (Gemini API): ~10M input + 1M output tokens	5 – 10
Cloud Storage, Secret Manager, Artifact Registry, Cloud Build	< 3
Firestore journal + Firebase Hosting (free tier at this volume)	< 1
Total running cost, single tenant	**≈ 120 – 160**
Lean pilot: 1 vCPU / 1 GiB, background jobs on Cloud Tasks, scale to zero	**≈ 35 – 55**

Indicative list prices (asia-south1); confirm with the Google Cloud Pricing Calculator. Excludes team effort. Cost does not grow with revenue: a 300-SKU catalogue adds roughly US$ 50–90 of Gemini calls a month on the same servers. Smaller brands can share one multi-tenant deployment.

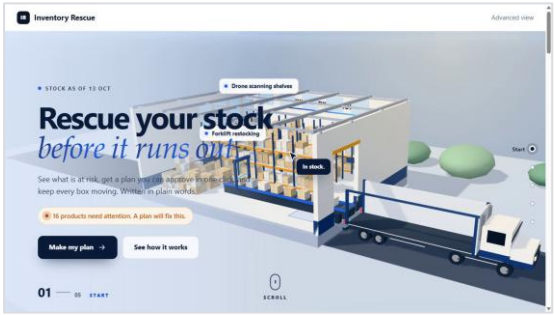
Usage assumptions

- Daily re-forecast of 30 SKUs (≈900 forecast calls a month).
- ≈300 plans, 600 messages and 1,000 assistant questions a month.
- Measured LLM outputs are small: 0.2–2.3k characters per call.

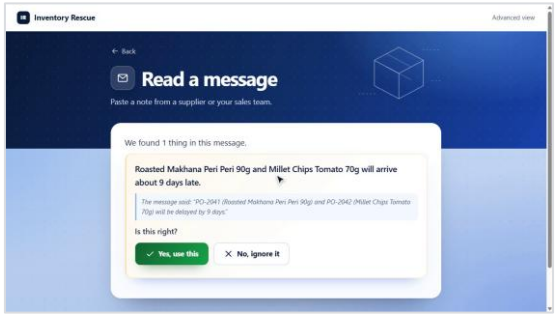
Cost vs value

- Dedicated deployment ≈ ₹10.5k–14k a month (at ₹88 / US$).
- Simulated loss avoided for the demo brand (₹4.3 crore revenue a month): ₹1.6–4.0 lakh a month.
- ≈ 11–38× the running cost. At a twentieth of this size (≈ ₹2.6 crore a year) a dedicated instance only breaks even, so small brands should share one.

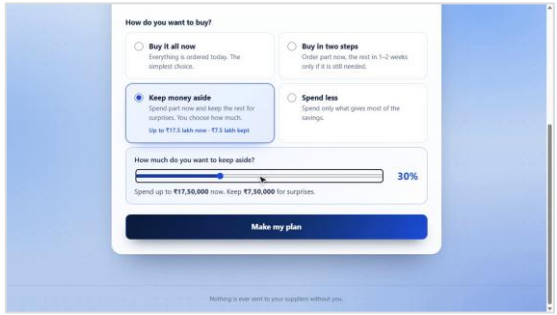
Snapshots of the prototype



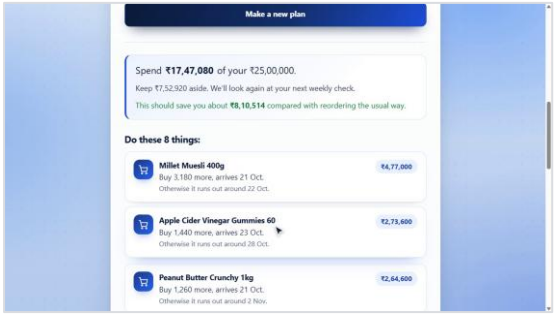
1. Opens on a 3D warehouse: what is at risk, in one line.



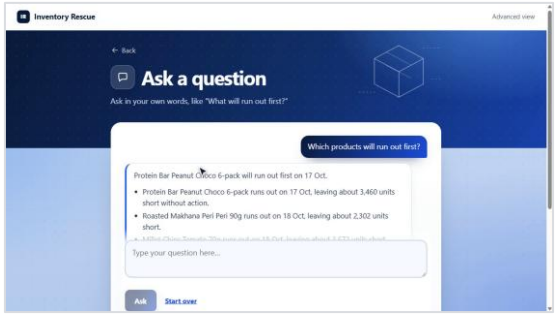
2. Read a message: Gemini quotes the line it used; you say yes or no.



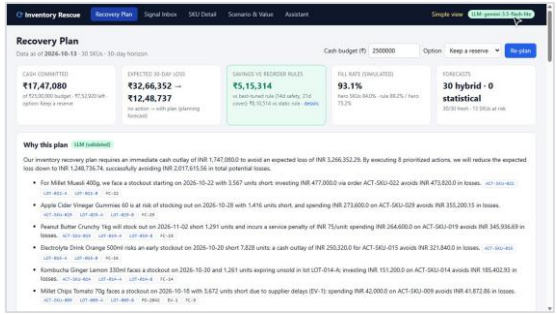
3. Type a budget; choose how to buy and how much to keep aside.



4. A plain to-do list: what to buy, by when, and the saving.



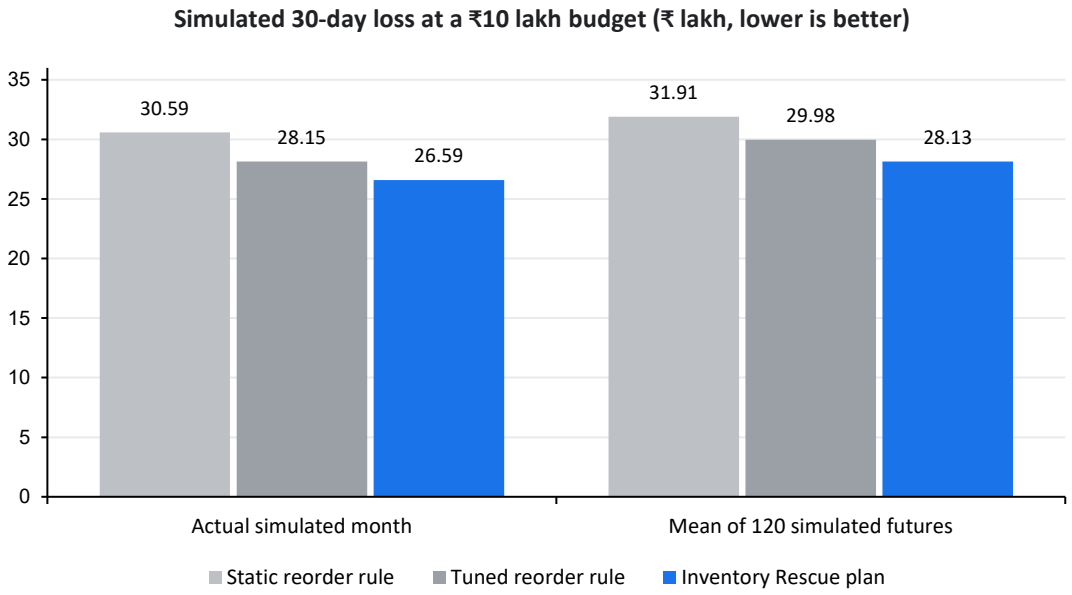
5. Ask in your own words; Gemini answers from checked numbers.



6. Advanced view: every number in the plan, with its evidence.

The Simple view is the default: a short 3D landing page, then three big buttons. Charts, robustness tests and scenarios stay one click away in the Advanced view.

Prototype performance report / benchmarking



Every policy starts from the same stock and budget and faces the same simulated demand and supplier delays. Fill rate 89.7% vs 88.6–88.8% for the rules; hero-SKU fill 80.3% vs 75.2%.

Engineering: 162 automated tests pass · 30 / 30 seed forecasts passed LLM validation · every LLM call logged with its status.

₹3.8 lakh / ₹1.8 lakh a month lower loss
Mean vs static / tuned reorder rule; better in 120 of 120 futures

30% reserve: ₹0.9–1.1 lakh further gain
Re-planned at weekly reviews; better in 10 / 10 calm and 9 / 10 shocked futures

Extraction F1 0.97 vs 0.65
LLM + validators vs keyword rules, 40 labelled messages; plans from LLM signals keep 100% of the gold-signal gain

Forecast WAPE 21.5% vs 23.0%
Hybrid vs statistical, 1,050 points (22.2% vs 23.9% on 16 text-only series)

Assistant: 6 / 6 answers number-checked
5 on the first try, 1 after one repair; 3–10 s each (gemini-3.5-flash-lite)

Gemini in the loop: what gemini-3.5-flash-lite adds

Task	With Gemini	Without the LLM	Guardrail
Read inbox messages into signals (40 labelled: email, WhatsApp, Hindi)	**F1 0.97; 95% of messages fully correct; 0 invented events**	Keyword rules: F1 0.65; 57.5% correct	Evidence quote must appear verbatim; SKU, date and quantity checked
Use those signals in a ₹10 lakh plan	**100% of the perfect-information gain (actual month and 40 worlds)**	Hand-labelled signals = 100%	A person confirms each signal before it is used
Tune event effects in the forecast (25 SKUs, 1,050 points)	**WAPE 21.5%; P10–P90 coverage 76%**	Statistical 23.0% / 75%; seasonal naive 24.1%	75/75 outputs passed validation; else statistical
Events known only from text (16 series)	**WAPE 22.2%**	Statistical 23.9%	Same validators
Planning assistant with tool calls (6 questions)	**6/6 answers number-checked (5 first try); median 4.0 s**	Labelled summary of the tool results	Every number must come from a tool result

Why gemini-3.5-flash-lite: fast (median assistant answer 4.0 s, tool calls included), every output passes the validators, and it fits the AI Studio free tier (15 requests a minute). Every call is logged with its status; a failed check falls back to a labelled non-LLM result.

Next: larger Gemini models (gemini-3.8-flash) on a billing-enabled key; the free tier allows 20 requests a day and one evaluation run needs about 130. Small sets: 40 messages, 75 forecasts, 6 questions, so gaps under about 0.5 pp WAPE are noise.

Additional details / future development

Product

- Multi-warehouse and channel allocation (marketplace vs D2C).
- Email / WhatsApp connectors and auto-drafted supplier follow-ups.
- SAP, Unicommerce and Shopify connectors for stock and POs.
- Plan for signal uncertainty: one misread day can move ₹1 lakh.
- Learn event multipliers from realised outcomes.

Platform on Google Cloud

- Live today on Cloud Run, Cloud SQL, Firestore and Firebase Hosting; next: HA and per-tenant data.
- Benchmark larger Gemini models on a paid key; route each task to its best model.
- Background jobs on Cloud Tasks; multi-tenant auth via Identity Platform.
- BigQuery ML forecasts as a second benchmark; Looker dashboards.

Validation & go-to-market

- Pilot with 2–3 mid-size FMCG / D2C brands on their real data.
- Measure realised savings against the simulated scorecard.
- Grow the gold set with real, operator-labelled messages.
- Pricing: subscription per SKU band, well below the savings.
- **Beyond FMCG:** pharma and cosmetics as-is; electronics, fashion and spare parts with expiry switched off.

Responsible AI: the LLM never sets quantities or money; every output is validated, labelled and logged, and signals need human confirmation.

Links

GitHub public repository
https://github.com/Xhahat/inventory-rescue

Demo video (3.5 minutes)
[Add demo video URL]

Final product link
https://inventory-rescue-511705.web.app · Firebase Hosting + Cloud Run (asia-south1)

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Thank you

